

HDOC - Day 12

First/Last Digit and Digit Comparison

Algorithm

1. Start. 2. Read positive integer n. 3. Display menu. 4. For option 1, obtain last digit using n%10; repeatedly divide a copy by 10 to obtain the first digit; display their sum and product. 5. For option 2, read digit k (0-9), extract each digit using %10, compare it with k and update lesser, equal or greater counters. 6. Display counts. 7. Repeat until Exit. 8. Stop.

Test Case Table

TC	Input(s)	Expected Output(s)	Actual Output(s)	Result
1	12345; choice 1	Sum=6, Product=5	Sum=6, Product=5	Pass
2	12345; choice 2; k=3	less=2, equal=1, greater=2	2, 1, 2	Pass
3	707; choice 1	Sum=14, Product=49	Sum=14, Product=49	Pass

C Program

```
#include <stdio.h>
int main() {
    int n, choice, temp, last, first, k, d, less, equal, greater;
    do { printf("Enter a positive integer: "); scanf("%d", &n); } while(n <= 0);
    do {
        printf("\n1. Sum and product of first and last digits\n");
        printf("\n2. Count digits lesser/equal/greater than k\n3. Exit\nEnter choice: ");
        scanf("%d", &choice);
        switch(choice) {
            case 1:
                last = n % 10; temp = n;
                while(temp >= 10) temp /= 10;
                first = temp;
                printf("Sum = %d\n", first + last);
                printf("Product = %d\n", first * last); break;
            case 2:
                do { printf("Enter digit k (0-9): "); scanf("%d", &k); } while(k < 0 || k > 9);
                temp = n; less = equal = greater = 0;
                while(temp > 0) {
                    d = temp % 10;
                    if(d < k) less++; else if(d == k) equal++; else greater++;
                    temp /= 10;
                }
                printf("Count_lesser = %d\nCount_equal = %d\nCount_greater = %d\n", less, equal, greater);
                break;
            case 3: break;
            default: printf("Invalid choice.\n");
        }
    } while(choice != 3);
    return 0;
}
```

Compilation, Execution & Testing

```
puppie@Anvikas-MacBook-Air Desktop % clang day12.c -o day12
puppie@Anvikas-MacBook-Air Desktop % ./day12
Enter a positive integer: 12345
Enter choice: 1
Sum = 6
Product = 5
```

```
puppie@Anvikas-MacBook-Air Desktop % ./day12
Enter a positive integer: 12345
Enter choice: 2
Enter digit k (0-9): 3
Count_lesser = 2
Count_equal = 1
Count_greater = 2
```

```
puppie@Anvikas-MacBook-Air Desktop % ./day12
Enter a positive integer: 707
Enter choice: 1
Sum = 14
Product = 49
```

Result: The menu-driven program was compiled, executed and tested successfully against the prepared test cases.